

Exp No: 26

TRANSACTION LEVEL MODELING (TLM) USING SYSTEMC

Program

```
#include <systemc>

#include <tlm>

#include <tlm_utils/simple_initiator_socket.h>

#include <tlm_utils/simple_target_socket.h>

using namespace sc_core;

using namespace tlm;

using namespace std;

// Target

struct Target : sc_module {

    tlm_utils::simple_target_socket<Target> socket;

    SC_CTOR(Target) : socket("socket") {

        socket.register_b_transport(this, &Target::b_transport);

    }

    void b_transport(tlm_generic_payload& trans, sc_time& delay) {

        int* data = (int*) trans.get_data_ptr();

        cout << "Target received: " << *data << endl;

        delay += sc_time(10, SC_NS);

    }

};

// Initiator

struct Initiator : sc_module {

    tlm_utils::simple_initiator_socket<Initiator> socket;

    SC_CTOR(Initiator) : socket("socket") {

        SC_THREAD(thread_process);

    }

    void thread_process() {
```

```

tlm_generic_payload trans;
sc_time delay = SC_ZERO_TIME;
int data = 555;
trans.set_data_ptr(reinterpret_cast<unsigned char*>(&data));
trans.set_data_length(sizeof(data));
cout << "Initiator sending: " << data << endl;
socket->b_transport(trans, delay);
wait(delay);
}
};

int sc_main(int argc, char* argv[]) {
Initiator initiator("initiator");
Target target("target");
initiator.socket.bind(target.socket);
sc_start();
return 0;
}

```

Output

Compile done. Starting run...

```

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```

Initiator sending: 555

Target received: 555

Done